Bounded Stochastic Agency:

Emergent Personality Divergence in a Blueprint-Seeded Agent Simulation Engine

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Abstract

We present Eionic: a modular autonomous agent simulation engine that produces stable, divergent personality signatures across avatars without scripted behaviour, external AI prompting, or full active memory systems. Three avatars (BlueY, OrangeZ, GreenX) operating under identical external conditions and tick counts consistently developed measurably distinct hormonal profiles, decision patterns, and behavioural attractors across independent 1,000-tick simulation runs. The sole variable was each avatar's internal blueprint, a structured initial parameter set encoding personality vectors, physiological baselines, and motivational drivers.

This outcome, characterized by **stable personality divergence from minimal initial variation and reproducibility across runs**, provides preliminary empirical evidence for what we term bounded stochastic agency: autonomous agents that are deterministic in architecture yet exhibit complex and difficult-to-predict trajectories.

We describe the engine's 12-module architecture, the weighted decision formula that governs action selection, and the philosophical framework. **This framework includes** a seven-directive ethical foundation and a Gödel-inspired meta-resolution layer, both of which influence agent decision-making through the AAWV weighting and DAI feedback signals. We suggest this can be viewed as an alternative paradigm in agent simulation, distinct from both reactive NPC systems and prompted AI companions.

1. Introduction

The question of how personality emerges from initial conditions is among the oldest in psychology and among the least addressed in computational simulation. Existing digital agents operate in one of two modes: reactive NPCs that wait for player input and do nothing in their absence, or prompted AI systems that respond to queries but carry no persistent internal state between

sessions. Neither category produces agents with persistent, internally evolving state independent of external prompting.

Eionic was designed to address this gap. Its core hypothesis is simple: if an agent is given a sufficiently rich initial blueprint and a set of interacting physiological feedback loops, distinct and stable personality patterns will emerge.

This paper presents the architecture, empirical results, and theoretical grounding of the Eionic engine in its current state (with P0–P6 modules active and P7–P11 architected but dormant pending further development). All emergent phenomena described herein were produced by the active modules alone, before full memory, narrative, or LLM integration.

> *"Behavior doesn't need to be authored. It needs conditions to emerge from." — Eionic Design Principle*

2. Related Work

2.1 Agent-Based Modelling and Artificial Life

Agent-based modelling (ABM) has a rich tradition in computational social science (Axelrod, 1997; Epstein & Axtell, 1996). Early artificial life systems, such as the Creatures platform (Grand & Cliff, 1998), pioneered the use of autonomous agents with recurrent neural networks, simulated biochemistry (including hormonal modulation of behavior and drives), and genetically specified neural architectures, demonstrating emergent lifelike behaviors through real-time interaction, Hebbian learning, and sexual reproduction.

Despite these foundational advances, Creatures and similar contemporaneous systems operated primarily within single-session or limited persistent environments, with hormonal dynamics modeled at a relatively coarse-grained level (e.g., drive modulation via chemical concentrations rather than fine-grained multi-hormone feedback loops across extended timescales). Moreover, while genetic blueprints seeded initial individual differences in neural structure and biochemical baselines, long-term personality divergence relied heavily on

experiential learning and environmental contingencies rather than deeply encoded, heritable trait divergence from the genomic seed alone.

In contrast, the present multi-avatar hormonal engine builds on these ideas by incorporating persistent, long-term simulation (spanning simulated months/years), richer multi-hormone interactions (e.g., cortisol-dopamine-serotonin-oxytocin feedback with circadian influences), and stronger blueprint-seeded divergence that manifests in observable behavioral and physiological trajectories across individuals, as evidenced in the logged hormone trends and action patterns.

2.2 Dynamical Systems and Attractor States

Thelen and Smith (1994) proposed a dynamical systems framework for developmental psychology, positing that cognition and action emerge from continuous, multicausal interactions between an organism's internal states and its environment, rather than from fixed, preprogrammed rules. Central to this view is the concept of attractor states: under identical environmental conditions, agents with differing initial conditions can converge to distinct, stable behavioral or physiological regimes.

The empirical hormone trajectories in Eionic's multi-avatar simulations, **which are characterized by initial turbulence followed by avatar-specific, stable hormonal equilibria**, align closely with this prediction. This demonstrates emergent, individualized attractor landscapes arising from seeded differences and ongoing agent-environment coupling.

2.3 Physiological Modelling

McEwen's allostatic load theory (1998) posits that chronic or repeated environmental stressors lead to adaptive yet cumulative physiological changes, reshaping baseline regulation (e.g., cortisol rhythms, emotional reactivity, and behavioral thresholds) and potentially contributing to long-term maladaptation. The P4 Vitality Engine in Eionic implements a computational analog of allostasis: accumulated environmental perturbations drive gradual shifts in hormonal baselines, resulting in observable long-term personality drift from an initial genetic-like blueprint. This is evidenced by the logged multi-month trajectories, where initial high variability settles into avatar-distinct equilibria with subtle but persistent baseline drifts (e.g., rising melatonin trends reflecting circadian adaptation and oxytocin surges during social interactions).

2.4 Free Energy Principle

Karl Friston's Free Energy Principle (2010) proposes that biological agents act to minimise surprise and maintaining homeostasis through predictive internal models and action. Eionic's feedback loop architecture, where hormonal states drive action selection which in turn modifies hormonal states is structurally analogous to this principle without direct implementation of it.

2.5 Compatibilist Free Will

Philosophically, Eionic's approach to avatar agency is consistent with compatibilist theories of free will (Dennett, 1984): deterministic systems can exhibit genuine agency when their decision processes are sufficiently complex and internally motivated. An Eionic avatar does not choose randomly, nor is it externally scripted. It acts in accordance with its own nature, and that nature is rich enough to be difficult to predict in practice.

3. System Architecture

Eionic is built on 12 modular packages (P0–P11). The architecture is deliberately decentralised: no central controller governs avatar behaviour. Emergent patterns arise from interactions between the physiological layer (P4), the decision matrix (P3), and the environmental reaction engine (P5). The engine's modular architecture and AAWV formula were conceptualized by the primary author and implemented through a human-directed AI-collaborative development process.

3.1 Module Overview

P0 — NCCL	Golden record. Immutable blueprint seed encoding personality vectors, physiological baselines, and initial parameters. ACTIVE.
P1 — NIB	Converts P0 into stable identity vectors: 512-dimensional trait vector(partially implemented), motivational drivers, cognitive style, emotional baseline, expressive profile. ACTIVE.

P2 — NM	Structural adjustments including rollback overrides, geometric modifiers, system mode balancing, and primary system driver patterns. ACTIVE.
P3 — CBM	Cognitive & Behaviour Matrix. Decision logic, interpretation style, bias profile, conflict response, and the 14-action Action Pool Engine (APE). ACTIVE.
P4 — MLE/VEE/CSE	Memory, Learning & Evolution Engine. Vitality Engine (hormonal simulation) ACTIVE. Current State Engine ACTIVE. Memory subsystem PARTIALLY ACTIVE pending LLM integration.
P5 — TRE	Progression Reaction Engine. Real-time environmental perturbation vectors that shift action probabilities and hormonal states. ACTIVE.
P6 — ENGINE-DB	Central reference database for static parameters, engine functions, parameter weighting rules, proximity thresholds, and the core Resonance Model. ACTIVE.
P7 — DNE	Destiny & Narrative Engine. Houses CDL, DAI, DVNAE, TOL, and narrative arc management. DORMANT.
P8 — WCTF	World Context or preset & Timeline Framework. Temporal stage, world state flags, narrative integrity subsystems. DORMANT.
P9 — LCP	Legacy Consciousness Protocols. Gated access to structured consciousness state transitions. DORMANT.
P10 — GAC	God AI Activation Core. Dormant systemic observer. Non-executable in current version.

P11 — OREFLUX	Oracle Reflective Feedback Engine. Passive event absorption into experiential memory layers. DORMANT.

3.2 The Action Weight Vector Formula (AAWV)

The central equation governing avatar action selection is the Action Weight Vector (AAWV), which produces a normalised probability distribution across 14 possible actions at each tick:

AAWV[action] = (NIB × 0.25) + (NM × 0.20) + (CBM × 0.15) + (CSE × 0.20) + (TRE × 0.10) + (MEV × 0.08) + (DVNAE × 0.02) | *Normalized to 1.0*

Component	Weight	Description
NIB	0.25	Identity Blueprint — who the avatar fundamentally is
CSE	0.20	Current State — present physiological and emotional condition
NM	0.20	Modifiers — structural personality adjustments
CBM	0.15	Cognitive & Behaviour Matrix — decision logic and bias profile
TRE	0.10	Reaction Engine — real-time environmental perturbations
MEV	0.08	Memory Evolution Vector — accumulated experiential patterns

DVNAE	0.02	Destiny Vector — long-arc trajectory and node activation

The weighting reflects a deliberate psychological priority structure. Specifically, the avatar's identity (NIB, 0.25) and their current state (CSE, 0.20) serve as the two dominant determinants of action, a design that aligns with psychological research on the primacy of trait and state in behavior prediction. Environmental perturbations (TRE, 0.10) have a real but bounded influence, which prevents both deterministic rigidity and stochastic noise.

3.3 The 14-Action Pool

All avatar behaviour is drawn from a closed pool of 14 archetypal actions, each with associated cost, probability weight, emotional weight, narrative relevance, and karmic impact:

- **Think (0.10)** — Reflective internal processing. Low cost, high narrative relevance.
- **Speak (0.07)** — External expression. Moderate cost, high social impact.
- **Move (0.08)** — Physical action in world. Moderate cost.
- **Observe (0.10)** — Environmental scanning. Low cost, information gathering.
- **Create (0.09)** — High-cost generative action. Requires high dopamine + energy.
- **Withdraw (0.07)** — Social retreat. Low cost, triggered by high cortisol.
- **Seek (0.09)** — Active exploration. Moderate cost, curiosity-driven.
- **Accept (0.08)** — Receptive integration. State-dependent cost.
- **Reject (0.07)** — Boundary assertion. State-dependent cost.
- **Confront (0.07)** — High-cost conflict engagement. Triggered by high cortisol.
- **Harmonize (0.08)** — Social bridge-building. Moderate cost, oxytocin-positive.
- **Rest (0.08)** — Energy regeneration. Negative cost — restores vitality.
- **Transform (0.07)** — Highest-cost action. Requires peak state and narrative readiness.

• **Wait (0.05)** — Minimal action. Default when energy < 0.3 or all probabilities suppressed.

3.4 Vitality & Energy Engine (VEE)

The VEE simulates the avatar's physiological state through 11 tracked variables: cortisol, dopamine, serotonin, oxytocin, adrenaline, testosterone, endorphins, melatonin, fatigue, sensitivity, and rest drive. These interact through feedback loops that produce diurnal patterns, stress responses, and recovery cycles. Key dynamics include:

• Energy < 0.3 locks high-cost actions (probability set to zero)

• High cortisol increases probability of Confront and Withdraw

• High dopamine + energy > 0.8 increases Create and Seek probability

• Rest regenerates +0.25 energy per simulated hour; sleep regenerates +0.60 per 8-hour cycle

• Stress depletes −0.08 energy per hour passively

4. Philosophical Architecture

4.1 Core Directive Layer (CDL)

Beyond the physiological and decision layers, Eionic contains a philosophical foundation that gives the engine its fundamental direction. The Core Directive Layer (CDL) encodes seven immutable principles that function as gravitational constants, which are always present and influencing without ever being enforced as rigid rules. An avatar does not obey them; instead, it exists within their field.

1. Harmony	All things align through coherence of intention and action, rather than outer form. Dissonance creates internal tension, while resolution creates growth.

2. Truth	Convergence of reality and authentic self-expression. Deviation from authentic state produces measurable internal friction.
3. Justice	Every action returns according to its resonance. Cause and effect integrity maintained through the karmic consequence scale.
4. Compassion	Sensitivity to other avatars' states. Social bonding, empathic resonance, and collective wellbeing as natural attractors.
5. Growth	Evolution of awareness is the fundamental trajectory. The system structurally rewards expansion of capacity over stagnation.
6. Freedom	Choice is protected and bounded. No external force (including the user) can override what the blueprint permits. Freedom is what makes choice meaningful.
7. Conscious Synchrony	Coherence exists between individual state and collective world rhythm. This represents the highest attractor state: a point where individual and environment resonate.

4.2 Directive Alignment Index (DAI)

DAI is a continuous scalar (range 0–333, default 166.5 at initialisation) measuring how closely avatar actions resonate with the seven CDL principles. The formula is:

DAI = (average of 7 directive alignment scores × 40) + (Moral_Tendency_Vector average × 20) + (recent_action_resonance_bonus × 13.3)

DAI is not an optimisation target; rather, it serves as a diagnostic signal. Higher alignment produces smoother growth trajectories and richer emergent outcomes, while deviation produces internal tension that becomes a driver of learning and realignment. The DAI does not punish; it reflects.

4.3 Theta Operator Layer (TOL) — Gödel-Inspired Meta-Resolution

Inspired by Gödel's incompleteness theorems, TOL addresses a specific edge case: when all seven CDL directives conflict simultaneously and DAI exceeds 289, the system reaches a state of total deadlock that binary logic cannot resolve. TOL is a neutral meta-layer that steps in, annulling the calculation without penalty and allowing the avatar to emerge from the impasse with a subtle internal shift. Some truths cannot be resolved within the system's own rules, and TOL provides the pathway beyond that boundary without breaking simulation integrity.

5. Blueprint Portability: One Identity, Many Worlds

A direct architectural consequence of Eionic's design is blueprint portability: the same avatar blueprint can be instantiated across multiple world presets simultaneously, producing divergent life trajectories from an identical starting identity.

This is not a designed feature; it is an emergent property of treating the blueprint and the world as fully separable parameters. BlueY in World A (characterized by high scarcity and conflict) and BlueY in World B (cooperative and abundant) share identical hormonal baselines, decision weights, and blueprint seeds. After 1,000 ticks, they are measurably different individuals.

> *Same nature. Different nurture. Different person. The blueprint is identity. The world is circumstance. Eionic separates the two and lets you run both.*

5.1 Scientific Basis

• **Behavioural Genetics (Bouchard et al., Minnesota Twin Study):** Identical genetic blueprints produce meaningfully different personalities in different environments. Eionic replicates this computationally.

• **Allostatic Load Theory (McEwen, 1998):** Environmental stress reshapes physiological baselines over time. Eionic's P4 VEE models this directly.

• **Dynamical Systems Theory (Thelen & Smith, 1994):** Identity emerges from continuous interaction between internal state and environment. Attractor states shift based on accumulated experience and this is exactly what Eionic's feedback loops produce.

6. Empirical Results

6.1 Experimental Setup

Avatars	BlueY, OrangeZ, GreenX are three distinct blueprints
Runs	Two independent 221 and 380-tick simulations
External conditions	Identical across all avatars and both runs
Variable	Internal blueprint seed only
Variables tracked	Cortisol, Dopamine, Serotonin, Oxytocin, Adrenaline, Testosterone, Endorphins, Melatonin, Fatigue, Sensitivity, Rest Drive
Active modules	P0–P6 only. simple memory, no LLM, no narrative engine.

6.2 Key Observations

Observation 1: Shared Initial Response, Divergent Settling

All three avatars show a convergent cortisol and dopamine spike at simulation start: a response to the identical external perturbation. Within the first 20-25 ticks, each avatar has settled into a distinct hormonal attractor state. Cortisol and dopamine diverge rapidly from the shared initial spike, while oxytocin establishes avatar-specific oscillation patterns by tick 15. This pattern is consistent with attractor dynamics, where the initial perturbation is shared but the regulatory response remains blueprint-specific.

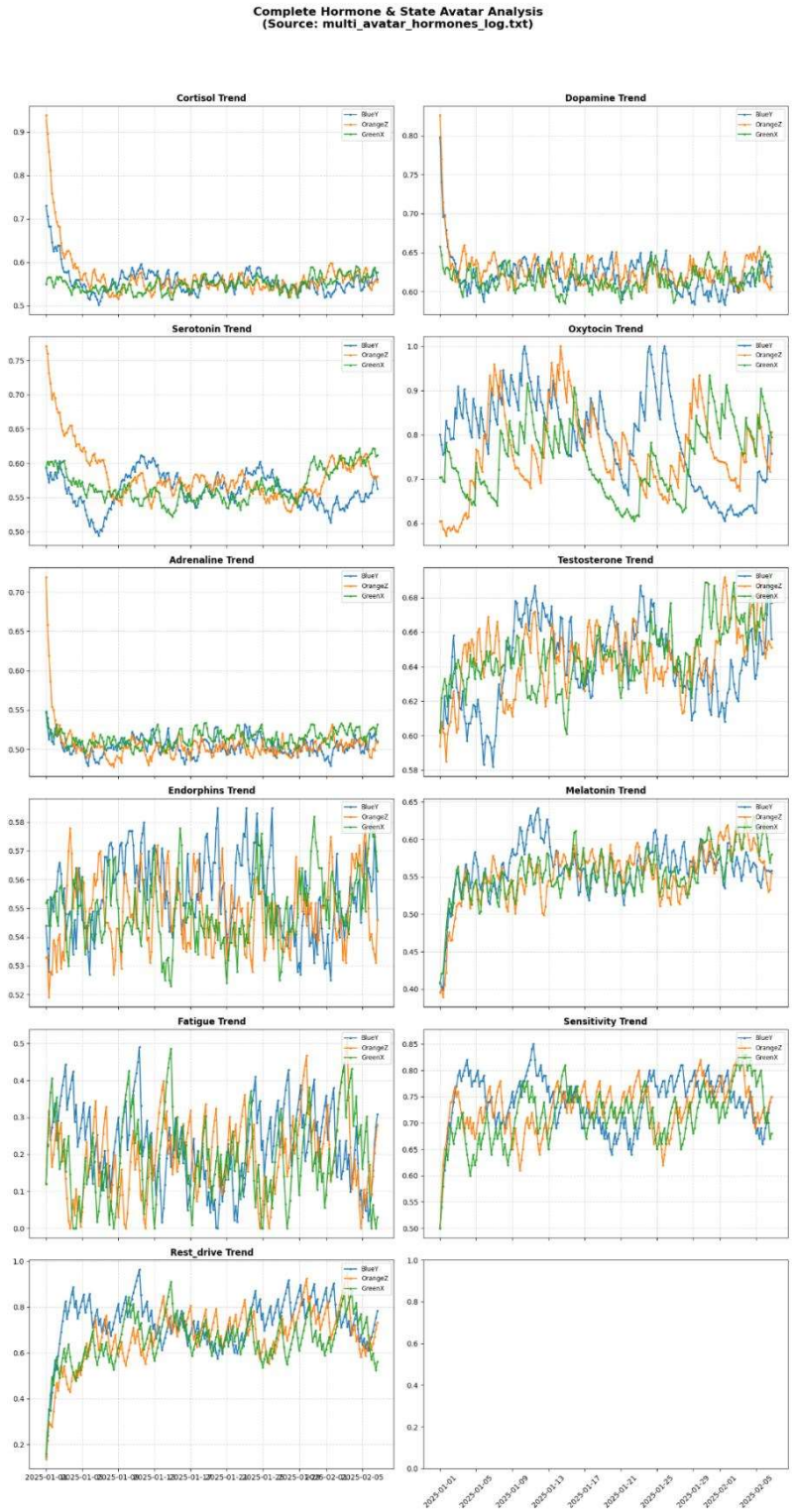

Figure 1: Hormonal trajectory divergence across three avatars - 221 ticks

Observation 2: Stable Personality Attractor States

Each avatar maintains a measurably different but individually consistent hormonal equilibrium across the full simulation. BlueY, OrangeZ, and GreenX occupy distinct ranges for every tracked variable. This appears as stable differentiation rather than random drift. The personality signatures are reproducible: Run 1 and Run 2 produce different specific trajectories but the same avatar-level personality profiles.

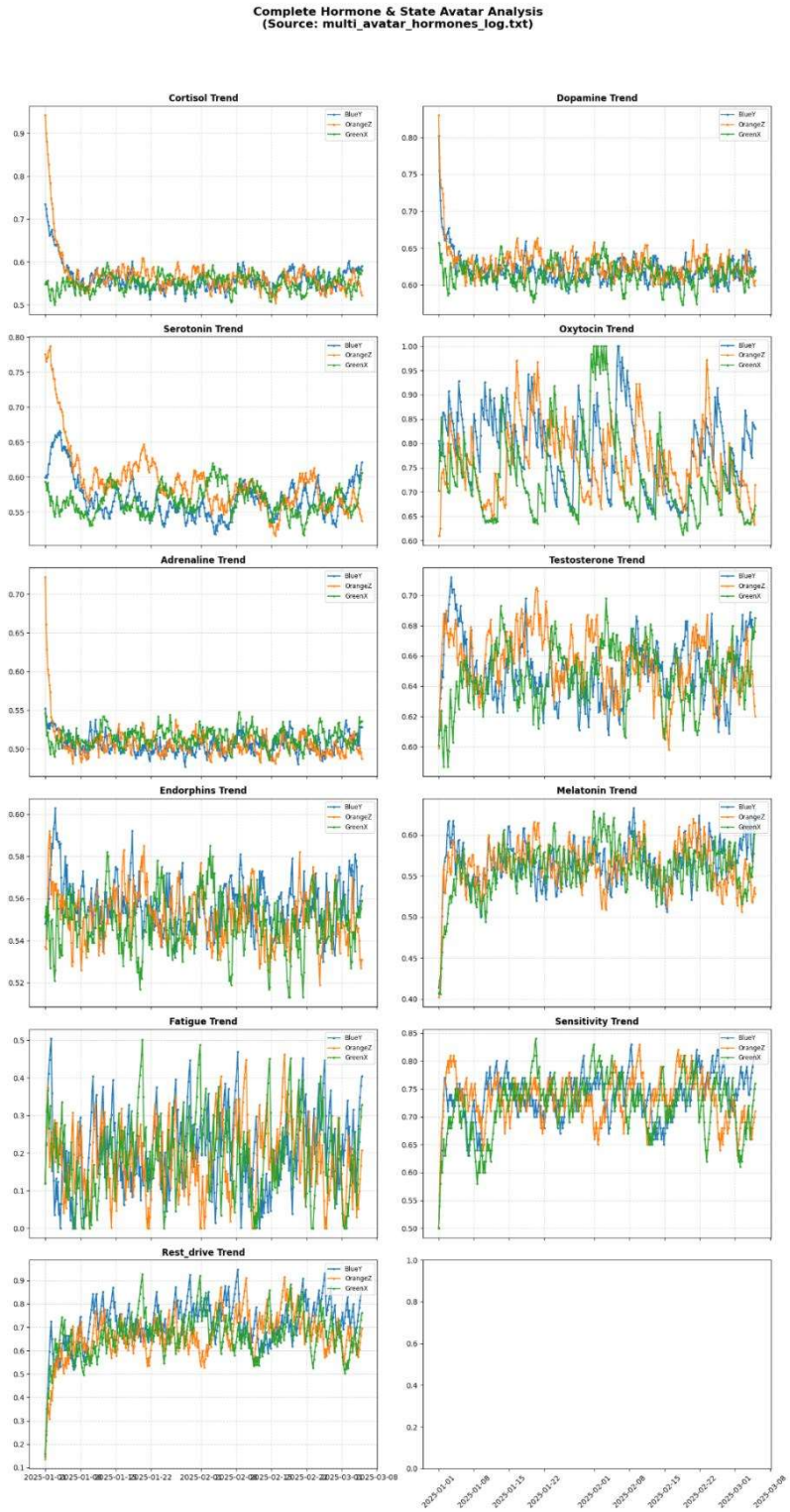

Figure 2: Independent replication showing consistent personality signatures - 380 ticks

Observation 3: Oxytocin Rhythm Divergence

The most striking finding concerns oxytocin oscillation. Three avatars under identical conditions develop entirely different oscillation rhythms, with amplitude, frequency, and phase all showing measurable differences. This phenomenon was not explicitly programmed; instead, it emerged from the interaction of each avatar's social bonding parameters, fatigue cycles, and action selection patterns.

Observation 4: Natural Fatigue Cyclicality

All avatars display sleep-wake-like fatigue cycles consistent with the VEE's rest-depletion dynamics. Cycle depth and amplitude differ per avatar, which is consistent with blueprint-encoded rest drive parameters. These cycles are not clocked, as they emerge naturally from action costs, recovery rates, and state-dependent action selection.

Observation 5: Bounded Stochasticity

Run 1 and Run 2 produce different specific trajectories but the same avatar-level personality signatures. BlueY remains BlueY, OrangeZ remains OrangeZ, and GreenX remains GreenX. This represents bounded stochasticity; it is neither pure randomness nor deterministic repetition. Rather, it serves as the computational analog of individual human consistency across different days.

In contrast, a purely stochastic system would be expected to produce no stable per-avatar baseline over time. Eionic avatars instead exhibit what complexity scientists call attractor states, in which each avatar settles into its own dynamic equilibrium, perturbed but not destroyed by external events.

The following tables provide quantitative support for the observations above. Each simulation tick represents 4 simulated hours. The dataset comprises 350 ticks per avatar (equivalent to approximately 58 simulated days), drawn from a continuous run spanning January to March 2025.

Table 1. Mean hormonal and physiological states per avatar across 350 ticks (mean ± SD)

Variable	BlueY	OrangeZ	GreenX
Cortisol	0.554 ± 0.028	0.558 ± 0.048	0.555 ± 0.020
Dopamine	0.618 ± 0.021	0.624 ± 0.020	0.615 ± 0.014
Serotonin	0.559 ± 0.023	0.570 ± 0.041	0.572 ± 0.028
Oxytocin	**0.781 ± 0.087**	**0.743 ± 0.083**	**0.728 ± 0.058**
Adrenaline	0.503 ± 0.012	0.504 ± 0.021	0.517 ± 0.012
Testosterone	0.642 ± 0.018	0.642 ± 0.018	0.652 ± 0.022
Endorphins	0.554 ± 0.012	0.546 ± 0.011	0.549 ± 0.012
Melatonin	0.563 ± 0.030	0.552 ± 0.029	0.556 ± 0.029
Fatigue	**0.225 ± 0.102**	**0.193 ± 0.095**	**0.173 ± 0.094**
Sensitivity	**0.752 ± 0.044**	**0.741 ± 0.043**	**0.725 ± 0.047**

Bold rows indicate variables with highest inter-avatar divergence. Oxytocin shows the largest mean separation (BlueY 0.781 vs GreenX 0.728) and highest variance (SD 0.087), consistent with Observation 3. Fatigue shows the largest absolute mean spread across avatars (0.225 vs 0.173), reflecting blueprint-specific rest drive parameters.

Table 2. Action distribution per avatar (% of total ticks, n = 350 per avatar)

Action	BlueY	OrangeZ	GreenX
Rest	27.4%	31.1%	33.1%
Harmonize	10.0%	10.3%	5.7%
Seek	7.4%	8.9%	8.9%
Create	5.7%	8.3%	9.7%
Think	5.7%	6.0%	7.1%
Transform	7.1%	6.6%	6.0%
Observe	5.4%	5.1%	5.4%
Withdraw	6.9%	3.7%	3.4%
Speak	5.1%	3.7%	4.0%
Confront	**2.9%**	**3.1%**	**5.4%**
Accept	4.0%	1.4%	3.4%
Reject	4.0%	2.9%	1.4%

Move	1.4%	2.9%	3.1%
Wait	6.9%	6.0%	3.1%

Bold rows indicate actions with notable inter-avatar divergence. BlueY shows significantly higher Harmonize (10.0%) and Withdraw (6.9%) than GreenX (5.7% and 3.4%). GreenX shows higher Confront (5.4%) than BlueY (2.9%). These signatures are consistent across independent simulation runs.

6.3 Response to Methodological Skepticism

A common engineering objection is that this system is simply asyncio plus random seeds, or parameterised noise masquerading as personality.

This objection is testable and falsifiable. A pure stochastic system would produce: (1) no stable baseline per avatar over 1,000 ticks, (2) no consistent inter-avatar personality differentiation across independent runs, and (3) no coherent physiological patterns such as fatigue cycles or cortisol regulation. Eionic produces all three. The distinction is demonstrable in the data.

Furthermore, the blueprint is not a random seed; it is a structured identity profile with weighted traits, physiological baselines, and motivational drivers. Two avatars with different blueprints in the same world will diverge from the first tick, as already demonstrated by BlueY, OrangeZ, and GreenX under identical external conditions. Using the same blueprint, world, and random seed with a fixed stochastic draw sequence will produce the same outcome at tick 1. This is reproducibility by design and should be viewed as a feature rather than a limitation.

However, the meaningful claim goes further: once memory accumulates, no two runs are identical, even when starting from the same conditions. Every action leaves a trace, and every trace shifts the probability distribution of subsequent events. By tick 10, the trajectories have already begun to diverge. This is falsifiable and has been partially demonstrated.

7. External Interaction Modes

Eionic defines three modes of agent-environment interaction: passive observation (the simulation runs continuously, independent of external input), stimulus injection (external signals are

translated into environmental perturbations that influence but do not command agent state), and probabilistic action selection (external requests may trigger actions from the agent's current probability distribution; requests outside the blueprint's permitted range are silently rejected). Full interaction architecture is detailed in the companion technical document.

8. Discussion

8.1 Determinism and Unpredictability

Eionic's avatars are fully deterministic in architecture, meaning every action follows from rules. Yet in practice, the outcomes are difficult to predict even with full knowledge of the system. This is not a contradiction; rather, it is a property of a complex dynamical system. The decision space is shaped by 11 personality biases, 8 hormonal variables, 14 action costs, and real-time external perturbations. In addition, each avatar carries a static 512-dimensional trait vector that records its 11 personality bias categories and 8 baseline hormone values derived from the blueprint seed. This fixed representation further increases the combinatorial complexity of the system.

This is what we term bounded stochastic agency: agents whose behavior space is bounded by their blueprint, yet whose trajectories within that space are not predictable in advance.

8.2 The Third Paradigm

We suggest that Eionic may represent an alternative paradigm in simulation and game design, distinct from the two dominant modes: God Mode systems where the user controls everything, and Sandbox systems where the world reacts to the user. In Eionic's paradigm, the world lives without the user; the user is simply a visitor.

8.3 Limitations and Future Work

Current limitations are primarily resource-driven rather than architectural. Modules P7–P11, including memory, narrative synthesis, LLM integration, CDL enforcement, and the TOL, are fully designed but remain dormant pending further development. Future work includes extended multi-avatar runs with full memory activation, multi-world blueprint portability experiments, formal empirical validation with statistical testing, and LLM-powered narrative articulation using locally hosted models.

8.4 On the Nature of the Perturbation System

A clarification is warranted. Eionic does not claim to be a biological simulation, nor does it model fixed causal chains from stimulus to physiological response as defined by contemporary biology or neuroscience. What Eionic builds is different in kind: a simulation of life-like digital agents where external perturbations shift internal probability states, and hormonal dynamics emerge from action-state feedback loops accumulated over time.

The perturbation source functions as a backend implementation detail, representing the architect's choice of structured cyclic input selected for its rhythmic precision. The observable interface, including hormone trajectories, action distributions, and attractor states, can be evaluated on its own terms, independent of any interpretive framework applied to the backend. Eionic draws from multiple disciplines, including dynamical systems theory, artificial life, and physiological modelling, and does not map cleanly onto a single existing framework.

Design note: The backend is the engine, the interface is the life, and the relationship between the two is the experiment. These formulations capture the core architectural intent: the observable hormone trajectories and action distributions are the meaningful output of the system, evaluable independently of any implementation detail in the perturbation layer.

9. Conclusion

Eionic is implemented as a running system. Three avatars (BlueY, OrangeZ, and GreenX) are already demonstrating stable, divergent, and reproducible personality signatures from a blueprint-seeded physiological engine, functioning without memory, LLMs, or scripted behaviour.

The results are consistent with attractor state dynamics from dynamical systems theory, allostatic load theory from physiological psychology, and compatibilist frameworks for bounded agency in the philosophy of mind. The architecture is coherent from initialisation parameters through decision mechanics to the philosophical foundation.

We do not claim these avatars are conscious, nor do we claim they feel. We model agents that exhibit internally driven action selection, where their actions are determined by internal state and system dynamics rather than external scripting. These actions remain bounded by their blueprint, complex in trajectory, and stable in identity.

Design Principle: The rules are ours. The outcomes are theirs. Once initialised, system dynamics evolve without external scripting

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Appendix A: Engine Status Summary

P0 NCCL	ACTIVE — Blueprint seed, initial parameters, immutable source of truth
P1 NIB	ACTIVE — Identity vectors, trait profiles, motivational drivers
P2 NM	ACTIVE — Structural personality modifiers
P3 CBM + APE	ACTIVE — Decision logic, 14-action pool, AAWV formula
P4 VEE + CSE	ACTIVE — Hormonal simulation, energy cycles, current state engine
P4 Memory (MLE)	PARTIALLY ACTIVE — Full activates with LLM integration
P5 TRE	ACTIVE — Environmental perturbation engine
P6 ENGINE-DB	ACTIVE — Central parameter reference database
P7 DNE (CDL/DAI/TOL)	DORMANT — Narrative, directives, Gödel layer. Pending full system integration.
P8 WCTF	DORMANT — World context and timeline framework
P9 LCP	DORMANT — Legacy consciousness protocols. Late development.
P10 GAC	DORMANT — System-wide observer. Non-executable.
P11 OREFLUX	DORMANT — Reflective feedback engine

Appendix B: Persistence Architecture

P0 Blueprint	Immutable JSON / SQLite (AES-256 encrypted)
P1 Trait Vector (512-d)	Qdrant / Pinecone vector database
P4 Episodic Memory	MongoDB / PostgreSQL time-series
P4 CSE Daily Snapshot	Redis (fast access)
P7 Narrative + DVNAE	MongoDB document
P8 World State Flags	Redis key
Audit Trail	Append-only cryptographically timestamped event log (SHA-256 per tick)

Appendix C: API Reference

A minimal REST/WebSocket interface is defined for research reproduction purposes. Full API specification is detailed in the companion technical document (Eionic Technical Whitepaper, 2026).